ECG Signal Denoising Using Digital Filtering and Wavelet-Based Methods

1. Student Information

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2. Project Overview

Electrocardiogram (ECG) signals are widely used to monitor the electrical activity of the heart and support the diagnosis of cardiovascular conditions. However, ECG recordings are often contaminated by different types of noise, including baseline wander caused by respiration or body movement, power-line interference at 50/60 Hz, and high-frequency noise caused by muscle activity or electronic devices. These disturbances may distort important waveform features such as the P wave, QRS complex, and T wave, reducing the reliability of further ECG analysis.

This project aims to design and evaluate an ECG signal denoising system using digital signal processing techniques. The proposed system will investigate conventional digital filters and a wavelet-based denoising method. We will measure their performance on quantitative measures such as signal-to-noise ratio (SNR), mean squared error (MSE) and root mean squared error (RMSE) in the future. The aim is to identify appropriate denoising methods that can be used to reduce the noise in ECG, while preserving the features of the ECG waveform. The quality of the signal to be analyzed in ECG studies is very important.

3. Background and Motivation

In practical ECG recordings, low-frequency baseline drift can affect the waveform, power-line interference can create narrow-band noise, and muscle activity can induce broadband high-frequency disturbances.If these noise components are not removed or reduced, they may interfere with feature extraction, heart-rate estimation, and automated ECG interpretation.

Digital filtering is a common method for ECG preprocessing because it is computationally efficient and relatively simple to implement. High-pass, low-pass, band-pass, and notch filters can be designed to suppress unwanted frequency components. However, excessive filtering may also distort useful ECG information.

Wavelet-based denoising provides an alternative method because wavelet analysis represents signals at multiple time and frequency scales. This makes it suitable for non-stationary biomedical signals such as ECG. By thresholding wavelet coefficients in this way noise level can be reduced and some of the important transient features, especially the QRS complex, can be preserved. This project was chosen as it combines basic signal processing concepts with an important biomedical application. It also provides a clear opportunity to compare the different methods with the measurable performance criteria.

4. Proposed Methodology

The project will be implemented mainly in Python with libraries like NumPy, SciPy, Matplotlib, PyWavelets, and WFDB. ECG recordings will be obtained from the MIT-BIH Arrhythmia Database available through PhysioNet.

The first stage will involve loading ECG recordings and visualising their time-domain characteristics. Frequency-domain analysis using the Fast Fourier Transform (FFT) may also be performed to identify important frequency components and different types of noise.

Controlled noise will then be added to selected ECG segments so that the denoising methods can be evaluated against a known reference signal. The noise types considered will include baseline wander, 50/60 Hz power-line interference, and high-frequency random noise.

Several denoising methods will be investigated. A high-pass or band-pass digital filter will first be used to reduce baseline drift and unwanted frequency components. A notch filter will then be designed to suppress power-line interference. In addition, a discrete wavelet transform (DWT) denoising method will be implemented. The ECG signal will be decomposed into wavelet coefficients, suitable thresholding will be applied to reduce noise, and the signal will then be reconstructed.

Different parameters, including filter cutoff frequencies, filter orders, wavelet families, decomposition levels, and threshold values, will be investigated. The processed ECG signals will be compared visually and quantitatively.

Performance will be evaluated using SNR improvement, MSE, and RMSE. The ability of each method to preserve important waveform features will also be examined by comparing the original, noisy, and denoised ECG signals. Finally, the filtering methods will be compared in terms of noise reduction, waveform distortion, implementation complexity, and computational performance.

5. Expected Outcomes

The expected outcome of this project is a working ECG signal denoising prototype that can load ECG recordings, apply different signal-processing techniques, display the original and processed signals, and calculate quantitative performance metrics.

The project is expected to demonstrate that different types of ECG noise require different filtering approaches. Conventional digital filters are expected to provide effective suppression of specific frequency components, while wavelet-based denoising may provide improved preservation of ECG waveform morphology.

The final deliverables will include Python source code, ECG data-loading instructions, signal visualisations, filter frequency-response plots, comparisons of different denoising methods, performance results based on SNR, MSE, and RMSE, and project documentation. A README file and demonstration code will also be provided in the public GitHub repository.

6. Timeline

Weeks 1–2: Finalise the project topic and scope, review ECG signal characteristics, create the GitHub repository, and identify suitable ECG datasets.

Weeks 3–5: Conduct the literature review, collect ECG data from PhysioNet, study common ECG noise sources, and determine appropriate evaluation metrics.

Weeks 6–7: Implement ECG data loading, signal visualisation, artificial noise generation, and basic digital filtering methods.

Weeks 8–9: Implement notch filtering and wavelet-based ECG denoising and conduct initial experiments.

Weeks 10–11: Optimise filter and wavelet parameters, calculate SNR, MSE, and RMSE, and compare the performance of the different methods.

Weeks 12–13: Complete the final evaluation, prepare figures and results, write the final report and README documentation, upload project materials to GitHub, and finalise the GitHub Pages site.

7. References

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